

**INTELLIGENT SINHALA SIGN LANGUAGE
COMMUNICATION PLATFORM WITH EMOTION
RECOGNITION AND ENVIRONMENTAL AWARENESS**

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BSc (Hons) degree in Information Technology Specializing in Information
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Declaration

I declare that the work presented here is original and that we have not plagiarized in any way, shape, or form any portion of this proposal into another work that has already been submitted for a degree or diploma in any other university or higher educational institution without acknowledgement.

Content that has already been written or published by someone else, unless the text specifically acknowledges it.

The supervisor and co-supervisor should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate dissertation under my supervision.

Signature of the supervisor

Date

Signature of the co-supervisor

Date

Abstract

Communication issues continue to be major obstacles for deaf, hearing-impaired, and speech-impaired people in Sri Lanka owing to the low level of awareness and use of SSL along with the unavailability of any comprehensive tool for communication. Most of the existing approaches have been limited to certain functions like learning the sign language and translating it into the written language but not both ways at once. To mitigate these weaknesses, this research proposes a smart Sinhala Sign Language Communication System where the process of recognizing emotions and awareness of the environment takes place together in a unified manner. There are four main aspects of the proposed smart system. To begin with, there is the aspect of Sinhala sign language recognition which recognizes hand movements, facial expression, and body movements and translates them into Sinhala text. Secondly, there is an adaptive learning system aspect which involves customized sign language learning based on artificial intelligence. Thirdly, there is the environmental sound recognition aspect where specific environmental sound classification such as sirens and horns is recognized and appropriate visual signals are generated for user safety purposes. Finally, there is two-way communication involving the use of visuals and sign language to facilitate interaction between children with speaking difficulties and their parents. This system is meant to improve communication efficiency, engage users in learning processes, and increase awareness about the environment among users. Through the use of artificial intelligence techniques in human-computer interaction methods, this study proves that there is scope for developing user-friendly assistive technology systems in Sri Lanka.

Keywords— Sinhala Sign Language Recognition, Emotion Recognition, Environmental Sound Classification, Adaptive Learning Systems, Human-Computer Interaction

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1. Introduction

1.1. Background Literature

Communication is a fundamental human requirement that enables individuals to express thoughts, emotions, needs, and intentions. However, for individuals with hearing and speech impairments, communication becomes a significant challenge in daily life. In Sri Lanka, Sinhala Sign Language (SSL) serves as the primary mode of communication for the deaf community, yet its usage outside specialized environments remains limited due to lack of awareness, inadequate technological support, and insufficient interactive learning systems. Similarly, children with speaking disabilities face additional challenges as they often depend on caregivers or simplified communication tools that do not fully support natural interaction.

With the rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), and Human-Computer Interaction (HCI), assistive technologies have evolved significantly. These technologies aim to bridge communication gaps and improve accessibility for people with disabilities. However, despite these advancements, most existing systems remain fragmented, focusing on isolated problems such as sign language recognition, basic learning applications, or simple alert systems. There is a lack of integrated solutions that address communication, learning, safety, and interaction in a unified manner.

Sign language learning systems have been widely developed in recent years using digital platforms and mobile applications. Most of these systems provide pre-recorded video lessons or gesture-based tutorials that allow users to learn sign language step by step. While these approaches are useful for beginners, they are static and do not adapt to the learner's progress. They also lack real-time feedback, meaning users cannot immediately verify whether a sign is performed correctly. Recent research in computer vision has introduced gesture recognition using deep learning models such as Convolutional Neural Networks (CNNs) and pose estimation techniques. These systems can identify hand gestures from images or video inputs; however, they typically focus only on recognition accuracy rather than personalized learning experiences. They do not consider learner behavior, memory retention, or adaptive curriculum design, which are essential for effective long-term learning.

In parallel, assistive technologies for hearing-impaired individuals have introduced environmental sound detection systems. These systems use mobile devices or wearable sensors to detect sounds such as alarms, doorbells, or notifications and alert users through vibration or visual cues. Although useful, most existing systems are limited in scope and do not differentiate between different types of environmental sounds. For example, they

may detect sound presence but fail to classify whether it is an emergency siren or a normal vehicle horn. Additionally, they do not prioritize alerts based on severity, which is crucial for real-world safety situations. Research in intelligent audio classification suggests that machine learning models can be used to categorize environmental sounds; however, such models are rarely integrated into real-time assistive mobile applications with structured alert systems.

Another major area of research is sign language recognition systems. These systems aim to translate sign language into text or speech using computer vision and deep learning techniques. Early systems relied on static image recognition, while more advanced approaches use video-based temporal modeling. However, most existing systems focus only on hand gestures and ignore other important components of sign language such as facial expressions and body posture. Studies in linguistics and sign language interpretation highlight that meaning in sign language is multimodal, where facial expressions can change sentence structure and emotional context, and body posture adds additional meaning. Despite this, many technological systems still rely on hand-only recognition, resulting in reduced accuracy and incomplete interpretation.

Augmentative and Alternative Communication (AAC) systems for individuals with speech disabilities have also been widely studied. These systems typically use symbol boards, image-based selection interfaces, or speech-generating devices to help users communicate. While AAC systems improve communication accessibility, they often support only one-way interaction, where the user expresses needs but does not receive natural conversational feedback. Additionally, most AAC tools are designed for general use and are not localized for specific languages such as Sinhala. Research in human-centered design emphasizes the importance of bidirectional communication systems, especially for children, where interaction should feel natural, engaging, and intuitive. However, current systems lack machine learning-based analysis to track communication development or predict user progress over time.

Based on the analysis of existing literature, it is evident that although multiple assistive technologies exist independently, there is no comprehensive system that integrates learning, communication, environmental awareness, and interactive support into a single intelligent platform. This research addresses this gap by proposing an Intelligent Sinhala Sign Language Communication Platform composed of four interconnected components.

The first component is an Adaptive Sinhala Sign Language Learning System that provides real-time gesture recognition using camera input combined with AI-based feedback mechanisms. Unlike traditional systems, it adapts to user performance, applies spaced repetition techniques, and personalizes the learning path based on individual progress.

This ensures that learners receive optimized training tailored to their strengths and weaknesses.

The second component is a Sound Alert System designed for hearing-impaired users. It continuously monitors environmental sounds using the device microphone and classifies them into categories such as vehicle horns and emergency sirens. It then generates prioritized visual alerts based on severity levels, ensuring that users are immediately informed of potential dangers in their surroundings.

The third component is a Sinhala Sign Language Reader, which addresses the limitations of existing recognition systems by incorporating multimodal analysis. It simultaneously processes hand gestures, facial expressions, and body posture to interpret Sinhala Sign Language more accurately. This enables real-time translation of sign language into text, allowing hearing individuals to understand communication without requiring prior knowledge of sign language.

The fourth component is a Sinhala Two-Way Communication System for children with speaking disabilities. This system uses a visual-based interface that allows children to communicate their needs through icons and sign selections. It also enables parents to respond using visual sign representations and text. Additionally, a machine learning module analyzes communication patterns and predicts the child's communication development level, providing valuable insights for parents and caregivers.

In conclusion, the reviewed literature highlights significant progress in assistive technologies but also reveals clear limitations in personalization, multimodal understanding, real-time interaction, and system integration. The proposed system addresses these gaps by combining multiple AI-driven components into a unified platform that supports learning, communication, safety, and inclusivity. This integrated approach not only enhances accessibility for deaf and speech-impaired individuals but also represents a step forward in the development of intelligent, human-centered assistive technologies.

1.2. Research Gap

Despite significant advancements in assistive technologies for deaf, hearing-impaired, and speech-impaired communities, several critical gaps still exist in current research and practical implementations, particularly in the context of Sinhala Sign Language (SSL) and localized communication systems in Sri Lanka.

One of the major gaps identified in existing literature is the lack of an integrated assistive ecosystem. Most current systems are developed as standalone solutions that address only a single problem area, such as sign language learning, gesture recognition, or sound detection. For example, learning applications focus solely on teaching signs, while recognition systems focus only on interpreting gestures, and alert systems focus only on environmental sounds. There is no unified platform that combines learning, real-time communication, environmental awareness, and caregiver interaction into a single intelligent system. This fragmentation reduces usability and limits real-world adoption, especially for users who require multiple assistive functions in daily life.

Another significant gap is the absence of adaptive and personalized learning in sign language education systems. Existing SSL or general sign language learning platforms typically follow a fixed curriculum structure where all users progress in the same sequence regardless of individual ability. These systems do not dynamically adjust difficulty based on learner performance, nor do they provide intelligent feedback tailored to individual mistakes. As a result, learners may either progress too slowly or struggle without targeted support. There is limited use of reinforcement learning, spaced repetition, or user behavior tracking to optimize long-term learning retention.

A further gap exists in multimodal sign language recognition systems. Most current research focuses primarily on hand gesture recognition using computer vision techniques. However, sign language is inherently multimodal, where meaning is conveyed not only through hand movements but also through facial expressions and body posture. Ignoring these additional modalities leads to reduced recognition accuracy and incomplete interpretation. There is still a lack of real-time systems that effectively combine hand, face, and body gesture analysis for Sinhala Sign Language in a unified model.

In the area of assistive safety technologies, existing sound detection systems show another important gap in intelligent prioritization and contextual awareness of environmental sounds. While some applications can detect basic sounds such as alarms or doorbells, they often fail to classify and prioritize critical real-world sounds such as emergency sirens versus normal traffic noise. Moreover, most systems do not provide tiered alert mechanisms that differentiate between high-risk and low-risk situations. This limits their effectiveness in real-world scenarios where immediate response to emergencies is crucial for user safety.

Additionally, there is a lack of bidirectional communication systems for children with speaking disabilities. Existing Augmentative and Alternative Communication (AAC) systems are largely one-directional, allowing children to express needs but not supporting natural conversational interaction with caregivers. These systems also lack engagement

features and do not provide meaningful feedback loops between child and parent communication. Furthermore, they are not culturally or linguistically optimized for Sinhala-speaking users, which limits their effectiveness in local contexts.

Another important research gap is the absence of machine learning-based behavioral analysis in AAC systems. Current tools do not analyze communication patterns over time to assess a child's development level or predict communication improvement. Without such intelligent tracking, parents and caregivers lack insights into progress, limiting their ability to provide targeted support or early intervention.

Finally, there is a broader gap in localized assistive technology for Sinhala Sign Language users. Many existing systems are developed based on international datasets and sign languages such as American Sign Language (ASL), making them less effective when applied to Sinhala Sign Language due to differences in grammar, structure, and cultural context. There is limited research focused specifically on Sri Lankan sign language datasets, real-world usage patterns, and culturally relevant interaction design.

In summary, the key research gaps identified include:

- Lack of integrated multi-functional assistive systems
- Limited personalization in sign language learning
- Insufficient multimodal recognition (hand, face, body combined)
- Weak prioritization in environmental sound alert systems
- One-way communication limitations in AAC systems
- Absence of behavioral analytics and ML-based progress tracking
- Lack of Sinhala-specific localized assistive solutions

Addressing these gaps is essential for developing a comprehensive, intelligent, and user-centered assistive communication platform. The proposed system directly targets these limitations by integrating learning, recognition, communication, and safety into a unified AI-powered ecosystem tailored for Sinhala Sign Language users.

1.3. Research Problem

Communication is an essential human right that empowers individuals to express thoughts, emotions, and needs. Yet for the deaf, hearing-impaired, and speech-impaired communities in Sri Lanka, this fundamental right is persistently constrained by systemic barriers rooted in language exclusivity, technological insufficiency, and social isolation. While global advances in Artificial Intelligence (AI), Machine Learning (ML), and computer vision have generated promising assistive tools, these innovations have largely failed to address the specific communicative, educational, and safety needs of Sinhala Sign Language (SSL) users in a holistic and locally contextualized manner.

The core research problem addressed by this project is the absence of a unified, intelligent, and culturally localized communication platform that simultaneously supports sign language learning, real-time sign language recognition, environmental sound awareness, and two-way communicative interaction for individuals with hearing and speech impairments in Sri Lanka. The existing landscape of assistive technologies is fragmented, narrow in scope, and largely developed for international sign languages such as American Sign Language (ASL), rendering them functionally inadequate for the SSL-speaking community.

One of the most pressing dimensions of this research problem is the fragmented nature of current assistive technologies. Available systems address isolated aspects of communication - sign language learning applications teach gestures but offer no real-time feedback; gesture recognition tools translate signs but ignore the full linguistic complexity of sign language; and sound alert systems detect audio events but lack contextual intelligence. No existing platform integrates these functions into a seamless, user-centred ecosystem. This fragmentation forces users to rely on multiple disconnected tools, creating cognitive load, reducing adoption rates, and failing to replicate the nuanced demands of real-world communication.

Sign language learning systems currently available in Sri Lanka and globally are predominantly static. They follow pre-defined, fixed curricula that do not account for individual learner performance, pace, or areas of difficulty. Users who progress too quickly or struggle with specific signs receive no differentiated support. This one-size-fits-all approach undermines effective acquisition of sign language and fails to leverage well-established learning science principles such as spaced repetition, adaptive difficulty scaling, and personalized feedback loops. The absence of real-time gesture verification further compounds this problem, as learners cannot confirm whether their signs are being performed correctly.

Sign language is an inherently multimodal linguistic system. Meaning is conveyed not solely through hand gestures but also through facial expressions, lip movements, body posture, and spatial positioning. Existing sign language recognition systems predominantly focus on hand gesture detection alone, neglecting the other linguistic dimensions that are critical to accurate interpretation. This reductionist approach results in significant gaps in recognition accuracy and leads to incomplete or erroneous translations. For Sinhala Sign Language specifically, which has its own grammar structure and expressive conventions distinct from international sign languages, this problem is further magnified by the near-complete absence of dedicated SSL datasets and recognition models.

For hearing-impaired individuals, awareness of the auditory environment is a matter of personal safety. Existing sound alert applications are largely reactive and context-blind - they may notify a user that a sound has been detected but fail to classify, prioritize, or respond intelligently to the nature of that sound. A vehicle horn in a parking lot and an emergency siren during a crisis both represent audio events, yet they carry vastly different risk levels. The inability of current systems to differentiate between types of sounds and deliver severity-tiered alerts constitutes a serious safety gap that leaves hearing-impaired users inadequately protected in dynamic real-world environments.

Children with speech and communication disabilities are disproportionately dependent on caregivers for expressing needs and receiving responses. Current Augmentative and Alternative Communication (AAC) tools facilitate one-directional expression - children can select symbols or icons to communicate, but the system does not enable natural, conversational reciprocity with parents or caregivers. Furthermore, these systems are not localized for Sinhala-speaking families, nor do they incorporate machine learning-based analytics to track communication development over time. The lack of developmental monitoring deprives parents and therapists of critical insights needed for early intervention and targeted support.

Taken together, these interconnected problems define a substantial and underserved gap in assistive communication technology for SSL-using communities in Sri Lanka. The absence of an integrated, intelligent, and locally contextualized platform that addresses learning, recognition, safety, and bidirectional communication in a unified system constitutes both the central problem statement and the primary motivation driving this research.

1.4. Research Objectives

The overarching aim of this research is to design, develop, and evaluate an Intelligent Sinhala Sign Language Communication Platform - a unified, AI-powered ecosystem that addresses the multifaceted communication, learning, safety, and interaction needs of deaf, hearing-impaired, and speech-impaired individuals in Sri Lanka. This aim is operationalised through a set of main objectives and corresponding specific objectives, each aligned to one of the four core system components.

1.4.1. Main Objectives

The following four main objectives frame the scope of the research and correspond directly to the four system components developed:

- To design and develop an Adaptive Sinhala Sign Language Learning System that delivers personalized, real-time sign language education through AI-driven gesture recognition, adaptive curriculum sequencing, and spaced repetition-based feedback mechanisms.
- To develop an intelligent Environmental Sound Alert System for hearing-impaired users that classifies real-world audio events in real time and generates prioritized, severity-tiered visual notifications to enhance situational awareness and personal safety.
- To build a Multimodal Sinhala Sign Language Reader that integrates simultaneous analysis of hand gestures, facial expressions, and body posture to achieve accurate, real-time translation of Sinhala Sign Language into text for hearing individuals.
- To create a Sinhala Two-Way Communication System for children with speaking disabilities that enables natural bidirectional interaction between children and caregivers, and incorporates machine learning-based communication development analysis.

1.4.2. Specific Objectives

Each main objective is further decomposed into specific, measurable objectives that guide the technical design and evaluation of the respective system components.

For the Adaptive Sinhala Sign Language Learning System, the specific objectives are:

- To implement a real-time hand gesture recognition module using Convolutional Neural Networks (CNN) and pose estimation techniques capable of identifying SSL gestures from camera input with high accuracy.

- To develop an adaptive learning engine that dynamically adjusts lesson difficulty, content sequencing, and review schedules based on individual learner performance and error patterns.
- To integrate a spaced repetition algorithm that optimizes long-term retention of sign language vocabulary by scheduling reviews based on memory decay models.
- To design an interactive feedback interface that provides immediate, corrective guidance when learners perform signs incorrectly, enabling self-directed improvement.

For the Environmental Sound Alert System, the specific objectives are:

- To develop a real-time audio classification model that distinguishes between different categories of environmental sounds, including vehicle horns, emergency sirens, doorbells, alarms, and ambient noise.
- To implement a severity-tiered alert mechanism that differentiates between high-risk audio events requiring immediate attention and low-risk events requiring routine notification.
- To design an accessible visual and haptic notification interface that delivers contextually appropriate alerts to hearing-impaired users in real time.

For the Multimodal Sinhala Sign Language Reader, the specific objectives are:

- To develop a multimodal recognition pipeline that simultaneously processes hand gestures, facial expressions, and body posture as integrated linguistic inputs for Sinhala Sign Language interpretation.
- To construct and curate a Sinhala Sign Language dataset encompassing multimodal features - including hand, face, and body data - to serve as the training corpus for the recognition model.
- To implement a real-time translation module that converts interpreted sign language inputs into accurate Sinhala or English text output for hearing users.
- To evaluate the recognition system against established benchmarks and compare performance with hand-gesture-only baselines to quantify the accuracy gains achieved through multimodal integration.

For the Sinhala Two-Way Communication System for children with speaking disabilities, the specific objectives are:

- To design a child-centered, visual interaction interface that enables children to communicate needs, emotions, and responses through icons, symbols, and sign representations in the Sinhala language context.

- To develop a caregiver response module that allows parents and guardians to reply using visual sign representations and text, enabling genuine conversational exchange.
- To implement a machine learning-based communication analytics module that tracks and analyses a child's communication patterns over time to assess developmental progress and predict improvement trajectories.
- To evaluate system usability and engagement through structured testing with children with speech disabilities and their caregivers in a Sri Lankan context.

1.4.3. Collective Objectives of the Integrated Platform

Beyond the component-specific objectives, the research is guided by three overarching platform-level objectives that apply across all four components:

- To design and integrate all four system components within a unified, cohesive mobile platform architecture that enables seamless access to learning, communication, recognition, and safety functionalities within a single application.
- To ensure that all system components are contextually localized for Sinhala Sign Language users in Sri Lanka, incorporating culturally appropriate interaction design, Sinhala language support, and SSL-specific datasets throughout.
- To evaluate the integrated platform against established usability, accessibility, and performance benchmarks to demonstrate measurable improvements over existing standalone solutions in accuracy, engagement, safety responsiveness, and communication effectiveness.

These objectives collectively position the proposed platform as a comprehensive response to the identified research gaps, advancing the field of intelligent assistive communication technology for Sinhala Sign Language communities in Sri Lanka.

2. Methodology

This research proposes an Intelligent Sinhala Sign Language Communication Platform that integrates four core components into a unified AI-driven assistive system: an adaptive learning module, a real-time sound alert system, a multimodal sign language reader, and a two-way communication system for children with speaking disabilities. The overall methodology follows a user-centered design approach combined with machine learning, computer vision, audio processing, and human-computer interaction (HCI) principles to ensure accessibility, personalization, and real-time responsiveness. The system is implemented as a mobile-based intelligent platform that uses device sensors such as camera and microphone to capture user inputs, which are then processed through AI models for recognition, classification, and prediction. All components share a common data management layer, allowing user interaction history, performance data, and behavioral patterns to be stored and reused for continuous system improvement. This enables the platform to adapt dynamically to individual users over time.

A key aspect of the methodology is multimodal data processing, where visual data (hand gestures, facial expressions, body posture), audio data (environmental sounds), and interaction data (user selections) are analyzed together to generate meaningful outputs. Machine learning models are used for gesture recognition, sound classification, and user behavior prediction, while feedback mechanisms ensure continuous learning and system optimization. The platform also follows a real-time processing framework, ensuring that outputs such as sign recognition results, sound alerts, and communication responses are generated instantly with minimal delay. In addition, an adaptive intelligence mechanism personalizes the user experience by adjusting learning content, alert priorities, and communication suggestions based on individual performance and usage patterns.

Methodology Summary Table

Aspect	Description
System Type	Mobile-based AI assistive platform
Core Approach	User-centered, adaptive, multimodal system
Input Sources	Camera (vision), Microphone (audio), Touch (UI interaction)
AI Techniques	Machine Learning, Computer Vision, Audio Classification
Processing Style	Real-time data processing and inference
Data Strategy	Shared user profile with continuous learning
Output Modes	Visual feedback, text, voice, and alerts

2.1. Adaptive Sinhala Sign Language Learning System

2.1.1 System Overview and Architecture

This study presents an Adaptive Sinhala Sign Language Learning System designed to enhance sign language education through intelligent personalization and real-time interaction. The system addresses limitations of traditional learning methods, such as static lesson delivery, lack of feedback, and absence of personalized learning paths. The proposed solution is implemented using client-server architecture, enabling efficient processing and real-time communication between components. The system consists of three primary layers.

Layer	Description
Mobile Application Layer	Captures gesture input and provides user interface
Backend Server Layer	Performs machine learning inference and adaptive decision-making
Data Storage Layer	Maintains user performance, progress, and learning history

The mobile application communicates with the backend via RESTful APIs, ensuring seamless data exchange and real-time feedback. This architecture allows the system to dynamically adjust learning behavior based on user performance.

2.1.1 Data Processing and Sign Recognition Model

The effectiveness of the proposed system largely depends on how accurately it can recognize Sinhala sign language gestures. To achieve this, the system follows a structured pipeline consisting of data collection, preprocessing, feature extraction, and deep learning-based classification.

Dataset and Data Collection

The system uses a custom-built dataset consisting of 25 Sinhala sign language letters, including both vowels and consonants. Data was collected using a camera-based setup from multiple individuals to capture variations in:

- Hand size and shape
- Gesture orientation
- Position relative to the camera

To ensure data quality, images were captured under controlled conditions:

- Plain background to reduce noise
- Consistent lighting to avoid shadows
- Multiple samples per sign for better generalization

This diversity allows the model to perform reliably even when different users perform the same sign in slightly different ways.

Preprocessing Techniques

Raw images captured from the camera cannot be directly used for model training or prediction because they may contain noise, inconsistent lighting, or irrelevant background information. Therefore, several preprocessing steps are applied to standardize the input data.

1. Contrast Enhancement

This step improves the visibility of the hand by increasing the difference between the hand and the background. It ensures that important features such as finger positions are clearly distinguishable.

2. Brightness Normalization

Images captured under different lighting conditions may vary in brightness. This step adjusts all images to a consistent brightness level, reducing the impact of environmental lighting changes.

3. Image Resizing

All images are resized to a fixed dimension (e.g., 224×224 pixels) to match the input requirements of the CNN model. This ensures uniformity and reduces computational complexity.

4. Feature Extraction (Hand Landmarks)

Using computer vision techniques, key points of the hand (such as finger joints and palm position) are extracted. These landmarks provide structural information about the gesture, improving recognition accuracy.

5. Multi-frame Processing (for Video Input)

Instead of relying on a single image, the system processes multiple consecutive frames when the user performs a gesture. This helps:

- Reduce prediction errors
- Handle slight movement variations

- Improve stability of recognition

Data Augmentation

To make the model more robust and adaptable to real-world scenarios, data augmentation techniques are applied during training. These include:

- Rotation (simulating different hand angles)
- Brightness variation (different lighting conditions)
- Slight distortions (real-world inconsistencies)

This artificially increases dataset diversity and prevents overfitting, allowing the model to generalize better when used in real environments.

Sign Recognition Model (CNN-Based Approach)

The core of the system is a Convolutional Neural Network (CNN), which is widely used for image classification tasks due to its ability to automatically learn spatial features.

Model Processing Steps:

1. Convolution Layers

These layers scan the input image using filters to detect low-level features such as edges, curves, and shapes. As the data passes through deeper layers, more complex patterns such as finger arrangements and hand structures are learned.

2. Pooling Layers

Pooling reduces the size of feature maps while preserving important information. This helps:

- Reduce computation
- Avoid overfitting
- Focus on dominant features

3. Fully Connected Layers

After feature extraction, the data is flattened and passed through fully connected layers, which learn relationships between extracted features and specific sign classes.

4. SoftMax Output Layer

The final layer produces a probability distribution across all possible sign classes. The class with the highest probability is selected as the predicted sign.

2.1.2 Adaptive Learning Using Reinforcement Learning

To provide personalized learning, the system incorporates a reinforcement learning (RL) framework that dynamically adjusts learning strategies based on user performance.

State Representation

The system maintains a learner profile using key performance indicators:

- Accuracy – Correctness of attempts
- Confidence – Certainty of prediction
- Response Time – Speed of performing signs
- Streak – Consecutive correct attempts
- Recency – Time since last practice
- Difficulty Level – Current learning stage

Action Space

Based on the learner's state, the system selects actions such as:

- Adjusting difficulty level
- Selecting appropriate feedback
- Scheduling review sessions

Reward Mechanism

The reward function considers:

- Accuracy improvement
- Learning progress
- Long-term retention
- User engagement

A Q-learning algorithm is used to optimize decision-making, enabling the system to learn the most effective strategies over time. This results in a fully personalized learning experience for each user.

2.1.3 Curriculum Design and Feedback Mechanism

The system employs a hybrid curriculum model that combines structured learning with dynamic adaptation.

Curriculum Structure

Learning is organized into progressive levels:

- Beginner: Basic vowel signs
- Intermediate: Consonants and combinations
- Advanced: Complex and dynamic gestures

Each level requires a minimum performance threshold before progression.

Dynamic Adaptation

The reinforcement learning model adjusts:

- Difficulty level based on user performance
- Practice frequency for weak signs
- Learning sequence for optimized progression

Feedback Mechanism

The system provides real-time, personalized feedback categorized as:

- Positive Feedback: Encourages correct performance
- Constructive Feedback: Provides improvement guidance
- Motivational Feedback: Enhances user engagement

Feedback selection is context-aware, ensuring relevance based on learner level and performance. This significantly improves learning effectiveness and user motivation.

2.1.4 System Implementation, Testing, and Evaluation

The system is implemented using modern technologies to ensure scalability and real-time performance:

- Frontend: React Native (mobile application)
- Backend: Flask (API and AI processing)
- Database: Stores user progress and interaction data

System Integration

All components are integrated using RESTful APIs:

1. User performs gestures via mobile camera
2. Input is sent to backend
3. CNN model predicts gestures
4. RL engine determines feedback and next action
5. Results are returned to frontend

This ensures smooth real-time interaction and adaptive learning behavior.

Testing Strategy

- Unit Testing – Validate individual components
- Integration Testing – Ensure proper interaction between modules
- System Testing – Evaluate overall system functionality
- User Testing – Assess usability and learning effectiveness

Evaluation Metrics

The system's performance is measured using:

- Recognition accuracy
- Response time
- User engagement
- Learning improvement over time

2.2. Environmental Sound Alert Module

2.2.1. System Overview and Architecture

This component presents an Environmental Sound Alert System designed to improve the real-world safety of deaf and hearing-impaired individuals through real-time audio classification and severity-tiered visual alerting. The system addresses the limitations of existing sound detection tools, which rely on simple volume thresholds and cannot distinguish between different sound types. The proposed solution uses a trained Convolutional Neural Network (CNN) to classify specific vehicle horn and emergency siren sounds, delivering prioritized alerts based on the danger level of each detected event.

The system is built on a client-server architecture consisting of three integrated layers:

Table 2.2.1: *System components, technologies, and responsibilities*

Component	Technology	Responsibility
Mobile Application	React Native / Expo	Records 2.5-second audio clips, dispatches them to the server, and renders severity-tiered visual alerts based on the response
Inference Server	Python / Flask	Receives audio clips, applies the preprocessing pipeline, runs CNN inference, and returns the predicted class, confidence, and severity level
Classification Model	TensorFlow / Keras CNN	Accepts a $(13 \times 40 \times 1)$ MFCC tensor and produces a seven-class softmax probability vector for sound classification

The mobile application records a 2.5-second audio clip every cycle and sends it to the Flask server via HTTP POST. The server processes the clip, runs inference, and returns a JSON response with the predicted class, confidence score, and severity level. The complete detection cycle from recording to alert display completes in 2.7 to 2.9 seconds, meeting the three-second latency target set in the research objectives.

2.2.2. Dataset Design and Collection

A custom dataset was assembled covering seven sound classes across two groups that are most relevant to hearing-impaired users in Sri Lankan Road environments.

Sound Class Selection

The vehicle horn group includes four classes:

- Bus horn
- Car horn
- Train horn
- Truck horn

The emergency siren group includes three classes:

- Ambulance siren
- Fire truck siren
- Police siren

A fifth candidate class, motorcycle horn, was evaluated but excluded after preliminary experiments showed significant spectral overlap with car horn recordings in the MFCC feature space, causing systematic classification confusion. Traffic ambient noise was also excluded as a class - since urban traffic is acoustically continuous, classifying it would trigger persistent false alerts in any outdoor environment.

Data Collection

Vehicle horn recordings were collected through field sessions at multiple Sri Lankan Road locations, varying:

- Distance from the sound source
- Background traffic density
- Time of day and acoustic environment

Emergency siren recordings were sourced from field recordings and publicly available audio repositories. The combined corpus contains approximately 3,230 clips. Siren classes have approximately 833 clips each while horn classes have approximately 182 clips each, reflecting greater online availability of siren audio.

Dataset Split and Class Imbalance

A stratified 64/16/20 train/validation/test split was applied to ensure proportional class representation. After quality filtering, 2,238 clips were retained. The 1:4.6 imbalance between horn and siren clips was corrected using scikit-learn's `compute_class_weight` function with the balanced strategy, assigning higher loss weights to minority horn classes during training.

2.2.3. Audio Preprocessing Pipeline

Raw audio files cannot be used directly for model training because they vary in sample rate, duration, and background noise content. A structured preprocessing pipeline standardises all input before feature extraction.

1. Loading and Resampling

All audio is loaded using librosa 0.10.0 and resampled to 22,050 Hz mono regardless of original source rate. This sample rate covers all acoustically relevant content in horns and sirens (up to ~11 kHz) while halving data volume compared to 44,100 Hz.

2. Fixed-Length Normalisation

Each clip is truncated or zero-padded to exactly 2.5 seconds (55,125 samples). This fixed length is a prerequisite for batched training and fixed-shape model inference.

3. Silence Trimming

Leading and trailing silence is removed using librosa's `effects.trim` with a 40 dB threshold. Clips trimmed below 0.3 seconds are reverted to the original to avoid information loss. This aligns active sound content toward the beginning of the window consistently across all samples. The same trimming is applied at inference time to maintain training-inference consistency.

4. MFCC Feature Extraction

Mel-Frequency Cepstral Coefficients (MFCCs) are used as the feature representation for three reasons:

- Data efficiency - a 55,125-sample waveform is compressed to a $13 \times 40 = 520$ -value matrix
- Perceptual grounding - the Mel scale approximates human auditory frequency resolution, separating perceptually distinct sounds in feature space
- Robustness - MFCCs are relatively insensitive to variations in recording volume and microphone response

13 MFCC coefficients are extracted across 40 time frames per clip, producing a (13, 40) matrix. Per-coefficient z-score normalisation is then applied using means and standard deviations computed from the training set and saved for reuse at inference time. The normalised matrix is reshaped to (13, 40, 1) to match the input format of Keras Conv2D layers.

2.2.4. CNN Model Architecture

A Convolutional Neural Network was selected as the classification model because the MFCC matrix has meaningful spatial structure in both the coefficient axis (spectral shape) and the time axis (temporal evolution). CNNs exploit this through shared convolutional filters, learning local acoustic patterns and combining them hierarchically into a class prediction. Recurrent architectures such as LSTMs were considered and rejected - since the full 2.5-second window is presented simultaneously, the task is holistic pattern recognition rather than sequential modelling.

Model Structure

The network consists of three convolutional blocks followed by a dense classification head:

- Convolutional Blocks ($\times 3$): Each block applies a Conv2D layer with 3×3 kernels and ReLU activation, followed by Batch Normalisation, MaxPool2D (2×2), and Dropout. Filter counts increase across the three blocks ($32 \rightarrow 64 \rightarrow 128$) to build progressively abstract feature representations. After three MaxPool operations, spatial dimensions reduce from (13, 40) to approximately (1, 5), producing a 640-dimensional flattened vector.
- Dense Head: Two fully connected layers of 256 and 128 units, each followed by Batch Normalisation and Dropout (0.50 and 0.40 respectively), before a seven-unit softmax output layer. Total trainable parameters: approximately 293,000.

Output and Confidence Gating

The softmax output produces a seven-element probability vector. The predicted class is the index of the highest value. A prediction triggers an alert only when the confidence meets or exceeds 85%, suppressing ambient noise misclassifications without requiring an explicit noise class in the dataset.

2.2.5. Training Configuration

The model was trained using the following configuration:

- Optimiser: Adam with an initial learning rate of 0.001. Adam's adaptive per-parameter learning rate handles the heterogeneous gradient magnitudes across MFCC coefficients naturally.
- Loss Function: Sparse Categorical Cross-Entropy, operating directly on integer class indices to avoid one-hot encoding overhead.
- Batch Size: 32, over a maximum of 100 epochs.

Three callbacks managed the training process adaptively:

- EarlyStopping - monitors val_loss with patience of 15 epochs; restores best weights on trigger
- ModelCheckpoint - saves the best val_accuracy checkpoint independently
- ReduceLROnPlateau - halves learning rate after 5 stagnant val_loss epochs, down to a minimum of 1×10^{-7}

Training ran for 76 epochs before EarlyStopping triggered. All training was conducted on CPU due to CUDA version incompatibility between TensorFlow 2.20.0 and the available GPU.

2.2.6. Severity-Tiered Alert System

Detected sound classes are mapped to one of three severity tiers based on their real-world danger profile, determining how the alert is presented to the user:

- Tier 1 - Emergency Sirens: Full-screen red overlay with maximum haptic intensity. Requires explicit user dismissal. Applied to ambulance, fire truck, and police siren detections.
- Tier 2 - Vehicle Horns: Prominent notification banner at the top of the screen, auto-dismissed after 3.5 seconds. Applied to bus, car, train, and truck horn detections.
- Tier 3 - Low Urgency: Subtle notification for informational events with no immediate danger implication.

All confirmed detections are logged in an alert history screen displaying the class label, confidence percentage, and timestamp, grouped by recency (Today, Yesterday, This Week).

2.2.7. System Testing and Evaluation

The system was tested across four levels to validate correctness, integration, and real-world performance.

Unit Testing - Preprocessing Pipeline

Five-unit tests validated the preprocessing pipeline:

- Resampling correctness across source rates of 8,000 Hz, 16,000 Hz, and 44,100 Hz
- Zero-padding behaviour for clips shorter than 2.5 seconds
- Truncation behaviour for clips longer than 2.5 seconds

- Silence trimming alignment verified using a synthetic clip with known silence boundaries
- Normalisation consistency between training and inference pipelines confirmed element-wise within floating-point tolerance

Model Inference Testing

Four model-level tests were conducted:

- Input-output shape consistency verified for tensor shape $(1, 13, 40, 1) \rightarrow (1, 7)$
- Softmax outputs confirmed to sum to 1.0 across all 448 test samples
- Per-class correctness tested on 10 held-out clips per class - all classified correctly
- Confidence scores confirmed to decrease monotonically under progressively degraded signal-to-noise conditions

Integration Testing - Flask API

The REST API is tested through automated HTTP requests:

- Health endpoint confirmed model-loaded status on startup
- WAV and M4A file inputs both classified correctly, verifying M4A fallback decoding
- Invalid input returned HTTP 400 with descriptive error message without server crash
- Concurrent requests handled correctly without race conditions on the cached model

End-to-End Mobile Testing

End-to-end testing is carried out using the React Native application on a physical Android device to verify the complete detection and alerting pipeline:

- Tier 1 full-screen alert is triggered using ambulance siren audio to verify the correct visual and haptic response
- Tier 2 banner behavior is tested with car horn audio to confirm it appears and auto-dismisses after the expected duration
- Ambient noise is recorded across multiple cycles to confirm no false positive alerts are generated
- Multiple detections are triggered sequentially to verify the alert history log captures the correct label, confidence, and timestamp for each event

Real-World Field Evaluation

Field evaluation is conducted in an active outdoor urban environment to assess system performance under realistic conditions:

- Detection latency is measured from audio recording start to alert display, targeting completion within three seconds
- The system is monitored during continuous ambient traffic noise to evaluate the false positive rate in real street conditions
- Sound sources are played at increasing distances to determine the effective detection range of the system in open air
- Overall classification accuracy is measured on the held-out test partition across all seven sound classes to evaluate model performance

2.4. Introducing a Sinhala Two-way Communication System for Children with Speaking Disabilities

This section outlines the methodology used in the design and implementation of the Sinhala Two-Way Communication System. This methodology combines the principles of HCI with those of machine learning in order to develop an efficient and intelligent software system.

The general approach is concerned with facilitating effective communication between children and their parents by using the visual medium of communication along with a behavioral prediction model that will aid communication analysis. This methodology is comprised of various stages such as requirements analysis, system design, data preparation, model creation, integration, and evaluation.

This component adds value to the research by covering the aspect of communication and interaction in the system.

2.4.1. Research Approach

The design process of this particular component will involve a mixed methodology, where both:

- User-centered Design (UCD) is used for the user interface development
- Data-driven machine learning is used for user behavior prediction

The chosen methodology guarantees that the system will not only satisfy technical requirements but also be in line with the requirements of the potential users.

The phases of the methodology include:

1. Requirement analysis
2. HCI system design
3. Datasets collection and processing
4. Machine learning models development
5. Integration of systems
6. Validation

2.4.2. Requirement Analysis

This involves analyzing the requirements of the principal users of the software: disabled children who cannot speak as well as their parents.

- Functional Requirements

These functional requirements have been established for the following purposes:

- Facilitate expressions of needs and emotions by the children through the use of visuals like icons and animations of signs
- Enable the parents to be able to understand the expressions made by the child
- Ensure that the parents can respond to these expressions via text/symbols
- Enable two-way instant communication
- Enable machine learning that will predict behaviors

➤ Non-Functional Requirements

The system should also meet certain qualities:

- High usability for the young user demographic
- Accessibility even without the ability to read
- No delay time in response during real-time interactions
- Easy to understand interface
- Secure processing of interaction information

2.4.3. HCI system design

The system design adheres to the principles of User Centered Design (UCD), where the requirements, abilities, and limitations of end users form the crux of the process throughout the development life cycle.

The processes involved in the design are:

- Identification of user attributes (age group, ability)
- Production of low fidelity prototype
- Production of high-fidelity prototypes through Figma
- Usability evaluation and improvement of the interface

HCI principles facilitate ease-of-use and clarity of the system for children who lack communication skills.

2.4.4. Datasets collection and processing

The machine learning section in the system is built upon an ordinal parent-based behavioral dataset, consisting of different behaviors observed in a child during his/her interactions and learning process.

Dataset Details

- No. of records: 200

- No. of attributes: 18
- Type of data: Ordinal (data values between 0 to 3)

The dataset contains details regarding the following:

- Ability to communicate
- Understanding of instructions
- Attention and involvement
- Learning capability
- Independence and confidence

The above attributes form a structured basis of the child's behavior.

2.4.5. Feature Engineering

In order to make the model more effective, we group the dataset in various categories. It will be easier to establish relationships between various behavioral traits.

Feature Categories

- Communication: The ability to communicate one's requirements and initiate interactions
- Instruction Comprehension: The ability to understand and obey instructions
- Concentration and Involvement: Time of focusing and task accomplishment
- Learning and Remembering: The ability to comprehend and remember concepts
- Independence and Self-confidence: The degree of independence and self-confidence

```
# --- Section 3: Configuration & Label Maps ---
FEATURES = [
    "age_group", "app_usage_bucket",
    "q1_familiar_instruction_response", "q2_new_instruction_response",
    "q3_response_speed_visual", "q4_basic_needs_expression",
    "q5_initiates_communication", "q6_multi_sign_combination",
    "q7_focus_duration", "q8_task_completion", "q9_reminder_dependency",
    "q10_sign_retention_next_day", "q11_repetition_needed",
    "q12_real_life_application", "q13_routine_understanding",
    "q14_routine_response_accuracy", "q15_independence_attempt",
    "q16_confidence_in_sign_usage",
]

TARGETS = [
    "communication_level",
    "instruction_understanding_level",
    "learning_retention_level",
    "attention_engagement_level",
    "independence_confidence_level",
]

TARGET_LABELS = {
    "communication_level": {0:"Emerging",1:"Developing",2:"Functional",3:"Strong"},
    "instruction_understanding_level": {0:"Limited",1:"Improving",2:"Consistent",3:"Advanced"},
    "learning_retention_level": {0:"Low",1:"Moderate",2:"Good",3:"Strong"},
    "attention_engagement_level": {0:"Very Short",1:"Short",2:"Stable",3:"Sustained"},
    "independence_confidence_level": {0:"Dependent",1:"Assisted",2:"Semi-Indep.",3:"Independent"},
}

TARGET_SHORT = {
    "communication_level": "Communication",
    "instruction_understanding_level": "Instruction",
    "learning_retention_level": "Learning",
    "attention_engagement_level": "Attention",
}
```

Figure 1: Features and targets

2.4.6. Data Processing

Preprocessing is done prior to training the model.

Some of the preprocessing tasks include:

- Conversion of ordinal values to numeric form.
- Data quality checking for missing values and inconsistencies.
- Partitioning of the data into training and testing set (80/20 ratio).
- Normalization of the data where necessary.

It should be noted that these tasks are carried out before feeding the model with data.

2.4.7. Machine Learning Model Development

In order to select the optimal model, several machine learning models are applied and analyzed.

Models Applied

- Random Forest
- Multilayer Perceptron (MLP)
- XGBoost (optimal model)

Each of the models is trained using the created data set.

XGBoost was finally chosen as the model based on its high level of performance and ability to work efficiently with structured data.

The strengths of XGBoost are the following:

- Accurate predictions
- High level of complexity of relationships that can be learned
- Robustness against overfitting
- Efficiency when dealing with small amounts of data

XGBoost operates by building an ensemble of decision trees in which each tree corrects the mistakes of the former ones.

2.4.8. Model Training and Testing Data

The data is split into training and testing data to assess the performance of the models.

- Training Data: 80% of the data
- Testing Data: 20% of the data

Models are trained on training data and tested using testing data to confirm their effectiveness.

2.4.9. Evaluation Metrics

Performance of the machine learning models will be evaluated based on evaluation metrics.

- Accuracy: Calculates the percentage of correct predictions
- F1-Score: Determines a compromise between precision and recall
- Confusion Matrix: Evaluates errors and distribution among classes

All of these metrics will give a complete evaluation of the model's performance.

2.4.10. System Integration

The last phase of the methodology is concerned with the incorporation of the machine learning algorithm within the HCI-driven framework.

These steps are included in the integration process:

- Incorporating the trained model in the backend
- Linking the user inputs to the predictor component
- Rendering the output via the user interface

3. Results and Discussion

3.1.Results

Adaptive Sinhala Sign Language Learning System

The Adaptive Learning System integrates a CNN sign classifier for 25 static Sinhala sign letters with a two-layer RL engine. The base `RLFeedbackAgent` uses a Q-table to adapt feedback granularity based on accumulated learner interaction history - providing detailed corrective cues to struggling learners and positive reinforcement to confident learners. The `AdaptiveLearningAgent` extends this with dynamic curriculum sequencing and spaced repetition scheduling derived from per-sign forgetting curve estimates, replacing fixed review intervals with individually predicted review priorities. The system operates in real time through a Flask REST API, with camera-based hand landmark detection handled by the MediaPipe pipeline and a 5-prediction smoothing buffer stabilising sign recognition output before RL-driven decisions are made.

Environmental Sound Alert Module

The Environmental Sound Alert Module was trained on approximately 3,230 audio clips across seven dangerous sound categories using a three-block CNN with 13×40 MFCC feature inputs. The model trained for 76 epochs before early stopping triggered, achieving a final training accuracy of 95.86% and a best validation accuracy of 94.40% at epoch 61. Evaluation on the held-out test set of 448 samples produced a top-1 test accuracy of ****92.19%****, with the three emergency siren classes (ambulance, fire truck, police siren) achieving the highest per-class performance due to their larger training representation (~833 clips each) and acoustically distinctive spectral signatures. The complete detection pipeline - from audio capture to mobile alert display - measured 2.7–2.9 seconds end-to-end over a local Wi-Fi connection. A confidence gate of 85% was found to be the optimal operating threshold, eliminating all background-noise false positives while retaining 97% of genuine hazard detections.

Smart Sinhala Sign Language Reader

The SSL Reader was trained on the signVideo dataset comprising 8,472 videos across 383 Sinhala sign classes - the largest vocabulary addressed by any published Sinhala Sign Language recognition system. Four model architectures were systematically evaluated: a Multimodal BiLSTM (~47% validation accuracy), a Multimodal Transformer (~41%), a Hybrid LSTM-Transformer (~52%), and the final Multi-Stream Fusion Model (~76.3%). The Multi-Stream Fusion Model outperformed the best baseline by 24 percentage points

while using approximately half the parameters ($\sim 2.52\text{M}$ vs $\sim 5.2\text{M}$), demonstrating that routing each modality (hand, face, pose) through a dedicated sub-network matched to its temporal characteristics is more effective than increasing model depth within a single shared architecture. Training converged at epoch 87, and evaluation on the held-out test set yielded a test accuracy of 80.1%. All 45 system test cases across feature extraction, augmentation, model inference, and API endpoints passed without failure.

Sinhala Two-Way Communication System

The Two-Way Communication System delivers a fully offline, dual-view mobile interface for non-verbal communication between children with speaking disabilities and their caregivers. The parent view provides a Sinhala Unicode text-to-sign conversion engine, a categorised icon library of 40+ culturally adapted icons across five domains, and 36+ pre-built daily communication templates across six scenarios (mealtime, school, health, play, bedtime, emotions). The child view provides an emoji-style emotion selector covering five states with intensity gradations and colour-coded feedback, a quick-needs button panel, and a conversation history log. All sign images, icons, and templates are stored locally on-device, ensuring complete functionality without internet connectivity - a critical design requirement for rural deployment in Sri Lanka. All six communication scenario domains were verified as fully covered during functional testing.

3.2. Research Findings

MFCC + CNN Is Sufficient for Real-Time Dangerous Sound Classification

The Sound Alert experiment established that a compact 13×40 MFCC feature matrix combined with a three-block CNN ($\sim 293,000$ parameters) is sufficient to classify seven dangerous sound categories at 92.19% test accuracy. Feature extraction requires only 5-10 ms on CPU and end-to-end latency measures 2.7-2.9 seconds - within the operational threshold for meaningful real-time warning. Per-coefficient z-score normalisation was identified as a critical preprocessing step: early experiments without normalisation showed significantly slower convergence because the large magnitude of the zeroth MFCC coefficient dominated gradients and suppressed learning in higher-order coefficients.

Modality-Specific Sub-Networks Substantially Outperform Single-Stream Architectures

The most significant finding of the SSL Reader research is that routing each input modality (hand gestures, facial expressions, body posture) through a dedicated sub-network

outperforms increasing model depth within a single shared architecture. The Multi-Stream Fusion Model (~2.52M parameters, 76.3% validation accuracy) outperformed the best single-stream baseline - the Hybrid LSTM-Transformer (~5.2M parameters, 52%) - by 24 percentage points while using approximately half the parameters. This demonstrates that the accuracy bottleneck in single-stream architectures is architectural mismatch, not insufficient capacity.

Transformer Architectures Are Ill-Suited to Low-Data Sign Language Tasks

Among the four SSL Reader architectures, the Multimodal Transformer achieved the lowest validation accuracy (~41%) despite being the second-largest model (~4.7M parameters). With approximately 22 training videos per class, the model lacked sufficient data to learn reliable global attention patterns across 383 sign classes. Architectures with stronger inductive biases - the LSTM's sequential gating and the TCN's local dilated convolutions - generalised better under the same data constraint. This finding has practical implications: inductive bias should be preferred over universal approximation when data is scarce in low-resource sign language contexts.

Skeleton-Level Augmentation Effectively Mitigates Data Scarcity

The SSL Reader experiment confirmed that skeleton-level data augmentation (seven stochastic operations: spatial rotation, scaling, Gaussian noise, temporal shift, time warping, frame dropping, and time masking) effectively trains deep models under severe data scarcity. Controlled ablation experiments showed an approximately 8-12 percentage point drop in validation accuracy when augmentation was removed. Horizontal flipping was explicitly excluded because Sinhala Sign Language distinguishes handedness - flipping a sign would change its meaning.

Attention-Based Fusion Provides Interpretable Modality Weighting

The attention fusion module in the Multi-Stream Fusion Model produces per-sample stream weights (w_{hand} , w_{face} , w_{pose}) that are interpretable by design. Analysis across the test set revealed linguistically consistent patterns: signs with strong distinctive handshapes showed high w_{hand} weights; signs with grammatically required facial components showed elevated w_{face} weights; signs involving body orientation showed elevated w_{pose} weights. This validates the architectural assumption that all three modalities contribute differentially, in agreement with sign language linguistic theory.

Confidence Gating Is the Key Usability Decision for Sound Alert Systems

The threshold analysis for the Sound Alert Module demonstrates that the confidence gate is the most impactful single engineering decision for an accessibility system of this type. At 60% confidence, three of ten background clips triggered false alerts - rendering the app

unusable. At 85%, all background clips were correctly suppressed while 97% of genuine hazard sounds were detected. At 95%, a small number of true positives at greater distances were incorrectly rejected. This finding suggests that for safety-oriented accessibility applications, the confidence threshold should be calibrated against the intended deployment acoustic environment rather than selected from general literature defaults.

RL-Based Personalisation Addresses Limitations of Static SSL Learning Apps

The Adaptive Learning component demonstrates that a Q-table RL agent integrated into a sign language learning app enables dynamic personalisation unavailable in conventional static-curriculum apps. The agent's ability to track per-sign proficiency, adapt feedback granularity, and schedule spaced repetition reviews addresses the two primary failure modes of static systems: learner frustration from excessive difficulty and disengagement from insufficient challenge. The spaced repetition component models the forgetting curve per sign, ensuring that signs at risk of being forgotten receive review at optimal intervals - a functionality absent from all currently published Sinhala Sign Language learning platforms.

Offline-First Design Is Essential for Rural Accessibility in Sri Lanka

The Two-Way Communication System's decision to store all sign images, icon libraries, and phrase templates locally reflects a critical finding about deployment context. Smartphone internet connectivity in rural Sri Lanka is inconsistent, and a communication tool for children with speaking disabilities that relies on server connectivity would be unusable in exactly the communities that most need it. The local database approach ensures full communication functionality regardless of network conditions, directly supporting the platform's equity objectives.

3.3. Discussion

Overall Platform Achievement

The Intelligent Sinhala Sign Language Communication Platform successfully realised four distinct, integrated components that collectively address the communication and safety challenges of the deaf and hearing-impaired community in Sri Lanka. Each component was technically validated against its objectives:

- The Adaptive Learning System operationalised RL-based personalisation for SSL education, providing the first published implementation of reinforcement learning for adaptive Sinhala Sign Language tutoring.

- The Sound Alert Module achieved 92.19% test accuracy on a 7-class dangerous sound classification problem, meeting the target accuracy of $\geq 90\%$ and the real-time latency requirement (< 3 seconds end-to-end).
- The SSL Reader achieved 80.1% test accuracy on 383 Sinhala sign classes - the largest vocabulary addressed by any published SSL recognition system - through a novel multi-stream architecture that outperformed all single-stream baselines by at least 24 percentage points.
- The Two-Way Communication System delivered a fully functional, offline-capable, family-centred visual communication interface for children with speaking disabilities, requiring no prior SSL knowledge from either communication partner.

Comparison of Adaptive Learning Results with Related Work

The Adaptive Learning System introduces RL-based personalisation to Sinhala Sign Language education - a combination that has no direct published precedent in the SSL context. Contextual comparison with the closest related systems situates this contribution:

Table : Adaptive Learning - Comparison with Related Systems

System	Language	Personalisation Approach	Spaced Repetition	RL Component
SignAll / Ace ASL (2021)	American SL	Rule-based difficulty tiers	No	No
SignGuru — Paudyal et al. (2020)	American SL	Error-driven feedback, fixed curriculum	No	No
Leitner System (classical)	General	Box-based card scheduling	Yes (fixed intervals)	No
This work — Adaptive Learning	Sinhala SL	Q-table RL agent, dynamic curriculum	Yes (ML-predicted intervals)	Yes

The two closest prior systems for sign language learning - Ace ASL (SignAll, 2021) and SignGuru (Paudyal et al., 2020) - both rely on hand-authored rule sets for feedback and difficulty adjustment. Neither system uses a learning agent that adapts its behaviour based on accumulated interaction history. SignGuru provides error-driven corrective feedback, which is conceptually aligned with this work's granular feedback mode, but its curriculum sequencing is static.

The classical Leitner spaced repetition system schedules reviews at predefined fixed intervals regardless of per-learner evidence. The Adaptive Learning System's spaced

repetition component replaces fixed intervals with ML-predicted review priorities derived from each individual learner's forgetting curve, enabling personalisation at a granularity that rule-based scheduling cannot achieve.

No published SSL learning system for any sign language variety uses a reinforcement learning agent as the curriculum and feedback controller. This work is, to the best of the authors' knowledge, the first implementation of RL-based adaptive learning for any South Asian sign language variety, and the first to combine RL-driven curriculum selection with ML-predicted spaced repetition in a single mobile-deployed learning system.

Contextualisation of SSL Reader Accuracy

The 80.1% test accuracy of the SSL Reader must be interpreted in proper context. For a 383-class classification problem, random chance accuracy is approximately 0.26% (1/383). The system achieves accuracy approximately 308 times higher than chance, demonstrating clearly that the model is learning meaningful, sign-specific representations. The 24 percentage-point improvement over the best single-stream baseline further confirms that the architectural contribution — not merely the data or parameter count — is responsible for the performance gain.

Comparison with related work supports the result's positioning:

Table : SSL Reader — Contextual Comparison with Prior Work

System	Language	Vocabulary	Accuracy
Kumara & Gunasekara (2019)	Sinhala SL	25 signs	85–95% (1 signer, controlled)
Jiang et al. (2021)	Chinese SL	100+ classes	85–90%+ (large corpus, many signers)
This work — SSL Reader	Sinhala SL	383 classes	80.1% (22 videos/class, limited signers)

The lower absolute accuracy compared to systems trained on large corpora is directly attributable to data scarcity (~22 videos per class) rather than architectural limitations. Published multi-stream skeleton-based SLR systems with hundreds of training samples per class achieve 85–90%+. With similar data volume, the Multi-Stream Fusion architecture is expected to reach comparable accuracy.

Comparison of Sound Alert Results with Related Work

The Sound Alert Module's 92.19% test accuracy on a 7-class custom dataset compares favourably with published benchmarks:

Table : Sound Alert - Comparison with Related Work

System	Dataset	Classes	Accuracy
Piczak (2015) — CNN	ESC-50	50	73.7%
Salamon & Bello (2017) — CNN	ESC-50	50	83.7%
Lim et al. (2017) — CNN	UrbanSound8K	10	79.0%
Park & Han (2019) — CNN	Custom (horn detect)	2	94.5%
This work — Sound Alert	Custom (Sri Lanka context)	7	92.19%

The closest comparable prior work - Park and Han's 2-class binary horn detection system - achieved 94.5% on a simpler problem. This work achieves 92.19% on a more challenging 7-class problem covering both horn types and emergency sirens, calibrated specifically to the Sri Lankan urban acoustic environment.

Comparison of Two-Way Communication Results with Related Work

The findings obtained using the suggested Sinhala Two-Way Communication System were compared with the results achieved in the case of other assistive communication systems in order to measure the efficacy and contribution of the proposed solution. These comparisons involve analyzing three main factors that include communication capability, usability, and intelligence.

As far as communication capability is concerned, most of the current systems in the field of assistive communication systems have been created with an emphasis on one-directional communication only. That means that such systems can perform tasks of translating signs into texts or vice versa but cannot offer two-way communication capabilities. On the contrary, the proposed system can be used in a two-way way communication allowing the child and the parent to communicate with each other at any moment of time.

Regarding usability, many systems do not pay special attention to making their interface suitable for young people with speech disorders. As a rule, such interfaces are based on texts which makes them difficult to use. However, the proposed system makes use of HCI technologies to develop its interface based on icons rather than text, offering easier navigation capabilities.

Yet another key aspect differentiating the system from other models is the inclusion of intelligent capabilities. Even though certain modern models employ machine learning algorithms to detect gestures or translate languages, such models often lack the ability to analyze behaviors associated with communication. In this case, the proposed system uses an XGBoost algorithm to predict the communication capability of users. The model demonstrated an impressive accuracy of 79%, thus proving to be effective in facilitating the operations of the system. Nevertheless, contrary to pure AI applications, machine learning in this case is applied only as a supportive tool in the system.

Another point differentiating the current solution from others is the adaptation of the latter to a particular language. As opposed to global applications, the proposed system is aimed at the Sinhalese language.

The comparison is summarized in the table below:

Aspect	Existing Systems	Proposed System
Communication Type	One-way	Two-way
Interaction	Limited	Real-time
Usability	Moderate	High (HCI-based design)
Intelligence	Minimal / Recognition-based	Behavioral prediction (XGBoost)
Localization	Generic	Sinhala-based
User Focus	General	Child–parent interaction

As seen from the above comparison, it is clear that the suggested system offers a better overall approach to the problem. While other existing systems try to solve certain parts of

the problem in terms of communication, none of them have attempted to integrate HCI, interaction, and artificial intelligence into one solution.

Limitations

Adaptive Learning System

The system faces a cold-start problem for new users with no interaction history. Initial lesson selection is effectively random until the RL agent accumulates sufficient evidence to personalise. Camera quality and lighting conditions affect hand landmark detection accuracy, which propagates into sign recognition quality and consequently feedback reliability.

Sound Alert Module

The training data was collected in a limited number of acoustic environments. In deployment, greater acoustic diversity may reduce accuracy. The model performs single-label classification per 2.5-second clip; simultaneous overlapping sounds are classified by the dominant spectral component only. The current system requires a Wi-Fi-connected Flask server; cloud deployment is required for wider usability.

SSL Reader

The primary limitation is data scarcity. With approximately 22 training videos per class, the model cannot fully learn within-class variability across 383 signs. The dataset contains recordings from a limited number of signers; multi-signer data would substantially improve robustness and generalisation. The system performs isolated sign recognition only — continuous signing recognition requires additional sequence modelling.

Two-Way Communication System

The current sign image library covers practical daily vocabulary but does not extend to specialised domains such as medical, legal, or technical contexts. Sign images are 2D drawings rather than video-based demonstrations, which may reduce comprehension accuracy for complex dynamic signs.

Integrated System Considerations

The four components share a common React Native mobile frontend and communicate through Flask REST API endpoints, forming a cohesive platform rather than four isolated tools. The integration architecture enables future cross-component synergies: for example, the SSL Reader's recognition capability could supply sign input data to the Adaptive Learning System, enabling real-time sign assessment without separate camera processing

pipelines. The Sound Alert Module's real-time monitoring can run concurrently with the Communication System, providing safety awareness during communication sessions.

The platform collectively addresses four dimensions of deaf accessibility in Sri Lanka that no single prior system has addressed simultaneously: (1) Safety awareness via Sound Alert, (2) Sign language recognition and communication via SSL Reader, (3) SSL education via Adaptive Learning, and (4) Family-centred non-verbal communication for children via Two-Way Communication.

3.4. Summary of Each Student's contribution

IT22315878 — Silva H.T.P.

Component: Adaptive Sinhala Sign Language Learning System

1. Sign Recognition Model Integration: Integrated the `sinhala_sign_language_classifier.keras` model (targeting 25 Sinhala sign letters) with a MediaPipe hand detection pipeline, including a 5-prediction smoothing buffer to stabilise recognition output. Implemented two hand detection backends (`cvzone HandDetector` and `raw MediaPipe Hands`) with a graceful fallback mechanism.
2. RL Feedback Agent: Designed and implemented the `RLFeedbackAgent` class using a Q-table reinforcement learning approach. The agent tracks per-learner, per-sign performance history and adapts feedback granularity - providing detailed corrective feedback for struggling learners and positive reinforcement for confident learners - based on learned Q-values.
3. Enhanced Adaptive Learning Agent: Developed the `AdaptiveLearningAgent` class extending the base RL agent with adaptive curriculum sequencing and spaced repetition scheduling. The agent maintains a proficiency state space and selects actions to maximize a composite reward signal combining assessment accuracy, long-term retention, and learner engagement.
4. Enhanced Preprocessing Pipeline: Implemented the `EnhancedPreprocessor` class incorporating CLAHE contrast enhancement, white balance correction, and optional background reduction for improved hand detection robustness under varying lighting conditions.
5. Video Sign Prediction Module: Built the `VideoSignPredictor` class to extend recognition from static signs to dynamic/video-based sign sequences, expanding the system's coverage beyond the 25 static letter signs.

6. Flask REST API (Adaptive Learning Backend): Implemented app.py as the unified Flask backend for the adaptive learning system, exposing endpoints for sign recognition, RL feedback generation, lesson sequencing, and learner progress tracking, serving the React Native mobile frontend.

IT22325464 — Kodithuwakku M.A.S.S.H.

Component: Environmental Sound Alert Module

1. Dataset Construction: Assembled and curated the 7-class audio dataset from locally collected vehicle horn recordings and publicly available siren recordings. Made and justified class-selection decisions - excluding motorcycle horn (acoustic overlap with car horn) and ambient traffic noise (continuous false-alert risk) - resulting in a focused dataset of approximately 3,230 clips.
2. Audio Preprocessing Pipeline: Designed and implemented the AudioPreprocessor class covering audio loading and mono resampling to 22,050 Hz, 40 dB silence trimming with short-clip fallback, fixed-length 2.5-second windowing, 13×40 MFCC feature extraction with Hann windowing and 128 Mel filterbanks, and per-coefficient z-score normalisation with training statistics saved to disk for consistent inference-time application.
3. CNN Model Design and Training: Designed the SoundClassifierCNN architecture — three convolutional blocks (32 → 64 → 128 filters, 3×3 kernels, BatchNorm, MaxPool, Dropout) followed by Dense(256) → Dense(128) → Dense(7, Softmax) — and implemented the full training configuration including per-class sample weights, EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint callbacks. Trained the model to 92.19% test accuracy on the held-out 448-sample test set.
4. Real-Time Inference Module: Implemented the RealtimeSoundDetector class supporting both microphone-based continuous monitoring and file-based inference, with result formatting that includes class name, severity level, confidence score, and human-readable display title.
5. Flask REST API: Built api_server.py to host the trained model, supporting WAV and M4A audio uploads from the React Native client, lazy model loading with in-memory caching, and structured JSON prediction responses containing class, severity, confidence, and timestamp.
6. Alert Tier System: Designed the three-tier severity alert framework: Tier 1 (full-screen red emergency overlay for ambulance, fire truck, police sirens), Tier 2 (auto-dismissing banner for vehicle horns), and Tier 3 (silent background history log).

7. System Testing: Conducted 18 documented test cases covering preprocessing unit tests, model inference validation, Flask API integration, end-to-end mobile pipeline testing, and real-world street environment testing.

IT22304674 — Liyanage M.L.I.S.

Component: SSL Reader (Smart Sinhala Sign Language Reader with Emotion Recognition)

1. Multi-Modal Feature Extraction Pipeline: Designed and implemented the complete feature extraction pipeline using MediaPipe Tasks API v0.10.21, integrating three landmark detection models in parallel (Hand Landmarker: 126-d, Face Landmarker with blendshapes: 232-d, Pose Landmarker: 99-d) into a 457-dimensional combined feature vector per frame, with a feature caching system to eliminate redundant extraction.
2. Root-Relative Normalisation: Designed and implemented stream-specific coordinate normalisation - wrist-centred for hand streams, nose-tip-centred for face landmarks, and mid-shoulder-centred for pose - with verified consistency between training-time and inference-time code paths to prevent distribution mismatch.
3. Skeleton-Level Augmentation Framework: Designed and implemented the SkeletonAugmenter and StreamSpecificAugmenter classes with seven stochastic operations: spatial rotation, spatial scaling, Gaussian noise, temporal shift, time warping, frame dropping, and time masking. Explicitly excluded horizontal flipping based on sign language handedness semantics.
4. Systematic Architecture Development: Designed and implemented all four model architectures in PyTorch - MultimodalLSTMModel, MultimodalTransformerModel, HybridModel, and MultiStreamFusionModel - and conducted systematic comparative evaluation, identifying modality-specific processing as the critical architectural factor.
5. Multi-Stream Fusion Model (Final Architecture): Designed the novel Multi-Stream Fusion architecture, including TCN sub-networks for hand and face streams (dilations 1, 2, 4), a BiLSTM sub-network for the pose stream with adaptive average pooling, and an attention fusion module projecting all streams to 512 dimensions with softmax-normalised stream weights. The model produces dual outputs: classification logits and interpretable per-stream attention weights. Achieved 80.1% test accuracy on 383 Sinhala sign classes - the largest vocabulary addressed by any published SSL recognition system.

6. Training Methodology: Implemented the MediaPipeTrainer class including cosine annealing with warm restarts ($T_0=30$, $T_{mult}=2$), Adam optimiser with weight decay, label smoothing cross-entropy ($\epsilon=0.1$), early stopping, and best-checkpoint saving. Managed the full 120-epoch training run converging at epoch 87.
7. Deployment and Integration: Implemented the Flask REST API (react_native_bridge.py) with /health, /labels, /predict_frame, and /predict_video endpoints, CORS support, threading lock for thread-safe model inference, and structured JSON responses. Solved Sinhala Unicode rendering on live camera frames using PIL with the Nirmala.ttc font.

IT22308870 — De Silva U.P.A.N.

Component: Sinhala Two-Way Communication System for Children with Speaking Disabilities

1. Devised a Sinhala Two-Way Communication System for children with speech impairments.
2. Created an interface for Human-Computer Interaction (HCI) which was centered around usability, accessibility, and ease of use for children ages 5-10 years old.
3. Created a communication interface for visual communication through the use of icons, sign animations, and less text so that children can communicate their feelings and needs.
4. Also, created a parent interaction interface where responses are sent through simple text messages as well as signs without any requirement for knowledge of Sinhala Sign Language.
5. Created a two-way interaction interface for real-time interaction between children and parents and better communication between them.
6. Applied concepts from User Centered Design (UCD) in developing the system to make sure that it is easy and user-friendly.
7. Used a machine learning algorithm called XGBoost to analyze behavioral data and classify communication-related features.
8. Selected the machine learning model XGBoost after testing its performance with Random Forest and MLP based on accuracy and F1 score.
9. Helped out in behavioral data analysis to identify features relating to communication, attention span, learning, and independence.
10. Localized the communication system so that it is only targeted towards Sinhala users.
11. Created the interaction part of the entire system.

12. Showed how an HCI system and AI work hand in hand, where machine learning is used to aid the communication process.

4. Conclusion

This research sought to solve the problem of poor communication faced by deaf and hearing-impaired people in Sri Lanka owing to poor knowledge about and lack of awareness of SSL. Current systems available in this field mainly concentrate on single features of their capabilities, such as learning of sign languages or unidirectional translation from sign language to natural language, making it impossible for such systems to provide a means of communication that allows users to interact with their environment, communicate emotions, etc. Therefore, this research project aimed at developing a comprehensive and intelligent SSL communication system that would combine advanced technologies within one platform.

The key novelty of this research project can be attributed to the fact that the SSL platform combines four distinct elements, namely recognition of signs, learning, perception of sounds in the environment, and two-way communication. While most systems available in this area are built separately, thus providing separate services, this platform ensures a comprehensive service that includes not only communication but also learning of new signs and safety of deaf and hearing-impaired persons when interacting with the environment.

The sign language recognition module proved its ability to recognize hand gestures, face expressions, and body movements and translate them into Sinhala text. With the use of the methods of multimodal deep learning, greater efficiency in recognizing the varied user input was observed. In this way, the effective use of modern artificial intelligence technologies was shown, including computer vision and sequence modeling.

The adaptive learning element is highly relevant to this system due to its capacity to provide individualized learning opportunities for users. By using an AI-powered feedback system and adaptive curriculums, the system will allow the development of users' competencies on different levels of proficiency, providing additional opportunities for knowledge acquisition and retention. The relevance of intelligent educational technologies was proven by this element.

The environmental sound alert module allows users to be notified about certain environmental sounds, which adds another layer of functionality to the system. With the use of this feature, users can become aware of any potentially dangerous situations, such as sirens, alarms, and car horns, through visual alerts. This functionality adds significant value to the system, being absent in many similar studies.

One of the major achievements of this project can be attributed to the creation of the two-way communication module that caters exclusively to children with speech disabilities and their parents. It involves the incorporation of images, animated sign language, Sinhalese texts, and emoticons to establish a realistic communication scenario. In terms of Human-Computer Interaction, this application places great emphasis on usability, accessibility, and culture-specificity, thereby making it functional and practical in the real-world context.

The entire system appears to be an excellent example of the effective utilization of key technological sectors. These include artificial intelligence, machine learning, data processing from multiple modalities, and human-centric interface design. This clearly indicates that the construction of such complex assistive systems is not only technologically feasible but also highly promising.

From an evaluation standpoint, user feedback and tests suggested that there was an increase in efficiency, user engagement, and situational awareness. The users were able to effectively interact through the use of the system, and because of the adaptive learning process, their continued use became necessary. Although the evaluation process was performed on a small scale because of time and resource constraints, the results obtained are highly encouraging in demonstrating the effectiveness and usability of the system.

Although the system is highly effective, some of the limitations that were observed included how lighting or background conditions could have an effect on the system's ability to detect gestures. Similarly, differences in signing styles between individuals could lead to recognition issues. The training set provided to train the model may not fully reflect real-world conditions.

As far as future research is concerned, the improvement of the system could involve the expansion of the database of users as well as the variety of environmental conditions under which they are recorded. Further advancement would allow for the integration of advanced deep learning techniques that will not only ensure the highest level of accuracy, but also the ability to process information in real time. Moreover, the use of mobile and cloud platforms can provide further flexibility as regards the use of the system.

However, the presented research also shows significant business prospects associated with the possibility of turning it into a useful assistive device for use in schools, therapy facilities, as well as at home for deaf people or those having speech disabilities. In other words, the system can be developed further into an actual mobile application or even a cloud-based one that would be used widely. Taking into account the growing popularity

of technologies aimed at inclusiveness, the presented invention can be easily scaled internationally, especially considering similar problems in other countries like Sri Lanka.

In conclusion, the current research has achieved the goal of designing a new, integrated, and human-centric system for supporting communication among the deaf and speech-impaired population in Sri Lanka. The focus on innovation, technology, users' needs, and feasibility highlights the possibilities of developing assistive devices that are based on both artificial intelligence techniques and human-computer interactions. In other words, the research adds a piece to the puzzle of creating a more open world through enabling communication.

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